

Name:.....

Student
Number

Teacher:.....



Pymble Ladies' College

Mathematics Extension 1

HSC Trial Examination

Term 3 2024

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks 70

SECTION 1 – 10 marks (pages 1-4)

- Attempt Questions 1-10
- Allow about 15 minutes for this section
- Answer each question on the multiple-choice answer sheet provided in the answer booklet.

SECTION II – 60 marks (pages 5-11)

- Attempt Questions 11-14
- Allow about 1 hours and 45 minutes for this section
- Answer each question in the appropriate space in the Answer Booklet. Extra writing pages are included at the end of each question.

SECTION I

10 marks

Attempt Questions 1-10

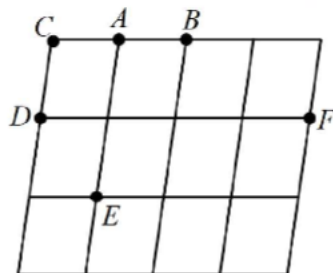
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

1. The grid shown is made up of identical parallelograms.

Let $\underline{a} = \overrightarrow{AB}$ and $\underline{c} = \overrightarrow{CD}$. What is the vector $\overrightarrow{EF}$ equal to?

- (A) $\underline{a} + 3\underline{c}$
 (B) $-3\underline{a} + \underline{c}$
 (C) $-3\underline{a} - \underline{c}$
 (D) $3\underline{a} - \underline{c}$



2. Four female and four male students are to be seated around a circular table. In how many ways can this be done if the males and females must alternate?

- (A) $4! \times 4!$
 (B) $3! \times 4!$
 (C) $3! \times 3!$
 (D) $2 \times 3! \times 3!$

3. Which of the following is equivalent to $\frac{d}{dx} \left(2 \sin^{-1} \frac{x}{2} \right)$?

- (A) $\frac{1}{\sqrt{1-x^2}}$
 (B) $\frac{2}{\sqrt{1-x^2}}$
 (C) $\frac{2}{\sqrt{4-x^2}}$
 (D) $\frac{1}{2\sqrt{4-x^2}}$

4. Which expression is identical to $2 \sin 3x \sin 5x$?
- (A) $-\cos 2x - \cos 8x$
(B) $\cos 8x - \cos 2x$
(C) $\cos 2x - \cos 8x$
(D) $\cos 2x + \cos 8x$
5. If $\sin A = t$ and $\cos B = t$, where $\frac{\pi}{2} < A < \pi$ and $0 < B < \frac{\pi}{2}$, then what is $\cos(B + A)$ equal to?
- (A) 0
(B) $\sqrt{1 - t^2}$
(C) $1 - 2t^2$
(D) $-2t\sqrt{1 - t^2}$
6. Let R be the region between the graphs of $y = 1$ and $y = \sin x$ from $x = 0$ to $x = \frac{\pi}{2}$. Which expression gives the volume of the solid obtained by revolving R about the x -axis?
- (A) $2\pi \int_0^{\frac{\pi}{2}} x \sin x \, dx$
(B) $\pi \int_0^{\frac{\pi}{2}} (1 - \sin x)^2 \, dx$
(C) $\pi \int_0^{\frac{\pi}{2}} \sin^2 x \, dx$
(D) $\pi \int_0^{\frac{\pi}{2}} \cos^2 x \, dx$

7. A spherical balloon is being inflated at a constant rate of $200\pi \text{ cm}^3 \text{ s}^{-1}$. At what rate is the radius of the balloon increasing when the radius is 10 cm?

- (A) 0.25 cm s^{-1}
 (B) 0.5 cm s^{-1}
 (C) 1 cm s^{-1}
 (D) 2 cm s^{-1}

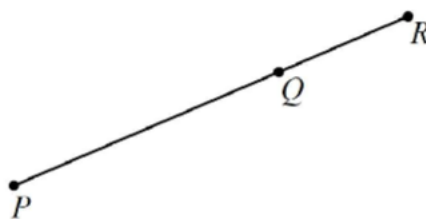
8. Which of the following is the domain of $y = 5 \cos^{-1}\left(\frac{2-x}{3}\right)$?

- (A) $x \in [1, 5]$
 (B) $x \in [-1, 5]$
 (C) $x \in [-5, 1]$
 (D) $x \in [-5, -1]$

9. PQR is a straight line and $PQ = 2QR$.

If $O\vec{Q} = 3\vec{i} - 2\vec{j}$ and $O\vec{R} = \vec{i} + 3\vec{j}$, where O is the origin, then what is $O\vec{P}$?

- (A) $-\vec{i} + 8\vec{j}$
 (B) $7\vec{i} - 12\vec{j}$
 (C) $4\vec{i} - 10\vec{j}$
 (D) $-4\vec{i} + 10\vec{j}$



10. The inverse function of $f(x) = \ln(x-1)$ is $g(x)$.

Which one of these statements must be true for all x in the domain of $g(x)$?

- (A) $g(x) < 0$
 (B) $g'(x) < 0$
 (C) $g''(x) > 0$
 (D) $g''(x) < 0$

SECTION II

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section.

Answer these questions in the Answer Book provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)	Marks
(a) Find the term independent of x in the expansion of $\left(3x^2 + \frac{2}{x}\right)^{12}$.	2
(b) Mrs Munro needs to decide the order in which to schedule 8 exams. Two of these exams are Mathematics Extension 1 and Mathematics Extension 2. Find the number of different ways that Mrs Munro can schedule the 8 exams so that Mathematics Extension 1 and Mathematics Extension 2 are NOT consecutive.	2
(c) Solve $\frac{x^2+10}{x} \leq 7$.	3
(d) The polynomial $P(x) = 8x^4 - 38x^3 + 9x^2 + ax + b$ has a double root at $x = 3$. Find the values of a and b , where a and b are real numbers.	3
(e) Evaluate $\int_0^2 \frac{dx}{\sqrt{16-x^2}}$.	2
(f) Use mathematical induction to prove that $8n^3 - 2n$ is divisible by 3 for all integers $n \geq 1$.	3

End of Question 11

Question 12 (14 marks)**Marks**

- (a) By using the substitution $t = \tan \frac{x}{2}$, solve

3

$$\cos x - \sqrt{3} \sin x + 1 = 0 \quad \text{for } 0 \leq x \leq 2\pi.$$

- (b) Use the substitution $u = x - 4$ to find the following integral:

3

$$\int x\sqrt{x-4} \, dx$$

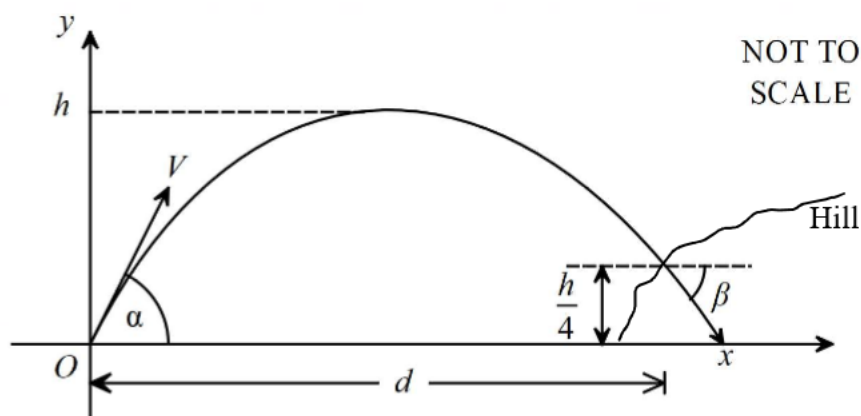
- (c) When the polynomial $P(x)$ is divided by $9x^2 - 1$ the remainder is $3x + 7$.

2

What is the remainder when $P(x)$ is divided by $3x + 1$?

Question 12 continues on page 7

- (d) Julie kicks a soccer ball from the origin O , which is on level ground, with velocity $V \text{ ms}^{-1}$, at an angle of α to the horizontal. The ball rises to a maximum height h and lands on a hill at distance d metres from the origin, with a height of $\frac{h}{4}$ metres, making an angle of β with the horizontal as shown.



Use the axes as shown and assume there is no air resistance.

The position vector of the ball, t seconds after being kicked, where g is acceleration due to gravity, is given by

$$\mathbf{r}(t) = (Vt \cos \alpha)\mathbf{i} + \left(Vt \sin \alpha - \frac{g}{2}t^2\right)\mathbf{j}$$

DO NOT prove this

- (i) Show that the maximum height h reached by the ball is

2

$$h = \frac{V^2 \sin^2 \alpha}{2g}.$$

- (ii) Show that the time taken for the ball to land on the hill is

3

$$t = \frac{(2 + \sqrt{3})V \sin \alpha}{2g} \text{ seconds.}$$

- (iii) Calculate the horizontal distance d travelled by the ball.

1

End of Question 12

Question 13 (15 marks)**Marks**

(a) Given the equation $f(x) = x \sin^{-1}\left(\frac{x}{2}\right)$, $-2 \leq x \leq 2$

(i) show that $f(x)$ is an even function

1

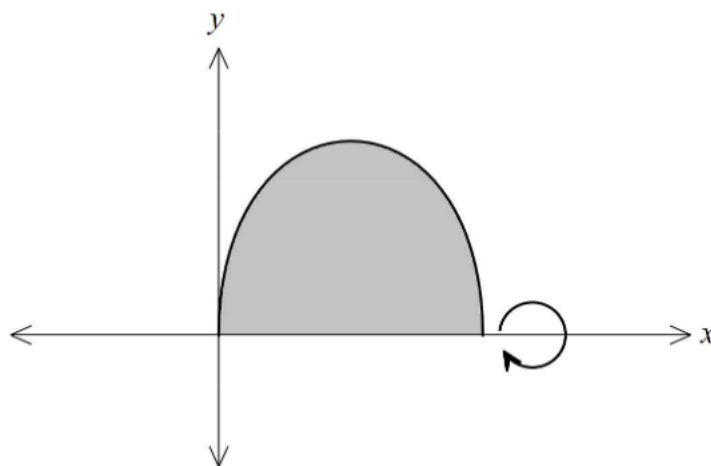
(ii) sketch the graph of $y = f(x)$ showing all features including intercept(s) and endpoints.

2

(b) After t minutes the temperature $T^{\circ}\text{C}$ of water in a jug is given by $T = 20 + 80e^{-0.2t}$. What is the rate at which the water is cooling when its temperature has fallen to half its initial value?

2

(c) Part of the graph of $y = \sqrt{\cos(3x)\sin(2x)}$ is shown in the diagram.



(i) By solving $\cos(3x)\sin(2x) = 0$, show that the smallest positive solution is $x = \frac{\pi}{6}$.

2

(ii) Hence, find the volume of the solid of revolution formed when the shaded region is rotated around the x -axis.

3

Question 13 continues on page 9

Question 13 continued.

Marks

(d) (i) Sketch the curve $y = -\tan^{-1} x$ and label the point where $x = 1$.

2

(ii) Find the area bounded by the curve, the x -axis and the line $x = 1$.

3

End of Question 13

Question 14 (16 marks)**Marks**

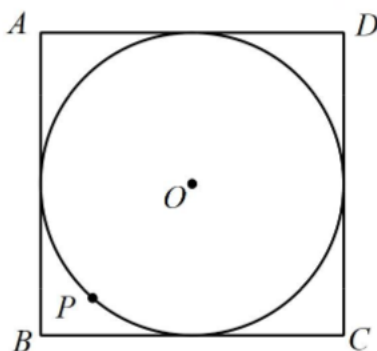
- (a) A bug moves such that its acceleration is given by $\frac{dv}{dt} = \sqrt{v+1} \text{ ms}^{-2}$. 3
Initially the bug is at rest. Find its velocity after 1 second.

- (b) A population of penguins on an island satisfies $\frac{dP}{dt} = 0.001P(400 - P)$ where P is the number of penguins and t is measured in years. Initially there are 50 penguins.

- (i) What is the carrying capacity of the island? 1

- (ii) Given that $\frac{1}{P(400 - P)} = \frac{1}{400} \left(\frac{1}{P} + \frac{1}{400 - P} \right)$, calculate when the population of penguins will reach 50% of the carrying capacity. 3

- (c) In the figure, the circle has centre O and radius r . The circle is inscribed in a square $ABCD$, and P is any point on the circle.



- (i) Show that $\overrightarrow{AP} \cdot \overrightarrow{AP} = 3r^2 - 2\overrightarrow{OP} \cdot \overrightarrow{OA}$. 2

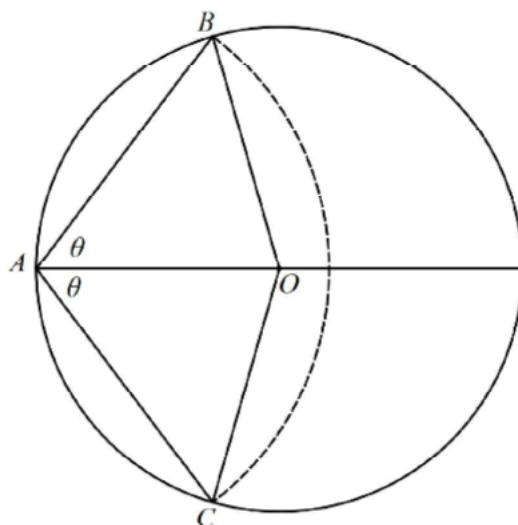
- (ii) Hence find $AP^2 + BP^2 + CP^2 + DP^2$ in terms of r . 2

Question 14 continues on page 11

Question 14 continued.

Marks

- (d) A is the point on the circumference of a circle with centre O and radius a . With A as the centre, an arc of radius r is drawn which meets the circle at two points B and C , and $r < 2a$. Arc length $BC = \ell$ and $\angle BAC = 2\theta$.



- (i) Show that $r = 2a \cos \theta$ and $\ell = 4a\theta \cos \theta$. **2**
- (ii) Hence, show that ℓ is a maximum when $\theta = \cot \theta$. **3**

END OF PAPER

Mathematics Extension 1
HSC Trial Examination
2024

Name:

Student Number:

Teacher's Name:.....

Section I – Multiple Choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A ○ B ● C ○ D ○

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ○ B ● C ○ D ○

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A  B  C  D 

- | | | | | |
|----|-------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1 | A ☐ | B ☐ | C ☐ | D ☒ |
| 2 | A ☐ | B ☒ | C ☐ | D ☐ |
| 3 | A ☐ | B ☐ | C ☒ | D ☐ |
| 4 | A ☐ | B ☐ | C ☒ | D ☐ |
| 5 | A ☐ | B ☐ | C ☐ | D ☒ |
| 6 | A ☐ | B ☐ | C ☐ | D ☒ |
| 7 | A ☐ | B ☒ | C ☐ | D ☐ |
| 8 | A ☐ | B ☒ | C ☐ | D ☐ |
| 9 | A ☐ | B ☒ | C ☐ | D ☐ |
| 10 | A ☐ | B ☐ | C ☒ | D ☐ |

SECTION II 60 marks

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the end of each question.

Question 11 (16 marks)

Marks

(a) ${}^{12}C_r 3^{12-r} (x^2)^{12-r} \left(\frac{2}{x}\right)^r$ *

2

$= {}^{12}C_r 3^{12-r} x^{24-2r} \times 2^r x^{-r}$

2 | Correct

$\therefore 24 - 3r = 0$

$r = 8$

Initial
expression

$\therefore$ Term independent of x is ${}^{12}C_8 3^4 \times 2^8$
 $= 10264320$

(b) Number of ways $= 8! - 7! \times 2!$

2

$= 30240$

2 | Correct

1 | 8! Correct

(c) $\frac{x^2+10}{x} \leq 7$



3

Consider

$x^2+10 = 7x$

Test $x = 1$

$\frac{1+10}{1} \leq 7$ False

$x^2 - 7x + 10 = 0$

$x = -1$

$\frac{1+10}{-1} \leq 7$ True

$(x-5)(x-2) = 0$

$x = 3$

$\frac{9+10}{3} \leq 7$ True

$x = 2, 5$

$x = 6$

$\frac{36+10}{6} \leq 7$ False

$\therefore x < 0$ or $2 \leq x \leq 5$

$\frac{x^2+10}{x} \leq 7$ $\times x^2$ $x \neq 0$

$x(x^2+10) \leq 7x^2$

$x(x^2+10) - 7x^2 \leq 0$

$x[x^2+10-7x] \leq 0$

$x(x-5)(x-2) \leq 0$



$x < 0$
or
 $2 \leq x \leq 5$

Question 11 continues on page 2

3 | Correct

2 | One inequality
Correct

1 | The beginning of
a correct approach

add
lines

Question 11 continued.

Marks

(d) $P(x) = 8x^4 - 38x^3 + 9x^2 + ax + b$ double root $x=3$ 3

$P'(x) = 32x^3 - 114x^2 + 18x + a$ 3 | correct

$P'(3) = 0 \therefore 32 \times 3^3 - 114 \times 3^2 + 18 \times 3 + a = 0$ 2. | Differentiated and one value correct

$a = 108$

$P(3) = 0 \quad 8 \times 3^4 - 38 \times 3^3 + 9 \times 3^2 + 108 \times 3 + b = 0$ 1 | Just differentiated

$b = -27$

(e) $\int_0^2 \frac{dx}{\sqrt{16-x^2}} = \pi/6$ ~~3~~ 2

2. | correct

$= \left[\sin^{-1} \frac{x}{4} \right]_0^2$ 1 | one error

$= \sin^{-1} \left(\frac{1}{2} \right) - \sin^{-1}(0)$

(f) Let the statement be $8n^3 - 2n$ is divisible by 3 $\forall n \geq 1$

3

When $n=1$ $8 \times 1^3 - 2 \times 1$
 $= 6$ which is divisible by 3

Assume the statement is true for $n=k$
 i.e. $8k^3 - 2k = 3P$ where $P \in \mathbb{Z}$ $k \geq 1$

ATP true for $n=k+1$
 i.e. $8(k+1)^3 - 2(k+1)$ is divisible by 3

Now

$$8(k+1)^3 - 2(k+1)$$

$$= 8(k^3 + 3k^2 + 3k + 1) - 2k - 2$$

$$= 8k^3 + 24k^2 + 22k + 6$$

From assumption $8k^3 = 3P + 2k$, substituting

$$= 3P + 2k + 24k^2 + 22k + 6$$

$$= 3P + 24k^2 + 24k + 6$$

$$= 3(P + 8k^2 + 8k + 2)$$

Since $k \in \mathbb{Z}$ and $P \in \mathbb{Z}$, $P + 8k^2 + 8k + 2 \in \mathbb{Z}$

$\therefore 8(k+1)^3 - 2(k+1)$ is divisible by 3

Hence, since the statement is true for $n=1$
 it is true for $n=1+1=2$ and so on by
 the principle of mathematical induction

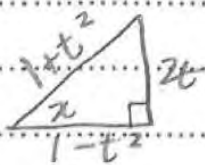
3	collect
2	Set up and part of argument
1	Just set up and conclusion

End of Question 11

Question 12

-1-

12) a) $t = \tan \frac{x}{2}$



$$\cos x - \sqrt{3} \sin x + 1 = 0 \quad 0 \leq x \leq 2\pi$$

$$\frac{1-t^2}{1+t^2} - \frac{2\sqrt{3}t}{1+t^2} + 1 = 0 \quad 0 \leq \frac{x}{2} \leq \pi$$

$$1-t^2 - 2\sqrt{3}t + 1+t^2 = 0$$

$$2 - 2\sqrt{3}t = 0$$

$$2\sqrt{3}t = 2$$

$$t = \frac{1}{\sqrt{3}}$$

$$\tan \frac{x}{2} = \frac{1}{\sqrt{3}}$$

$$\frac{x}{2} = \frac{\pi}{6}$$

$$x = \frac{\pi}{3}$$

check: $x = \pi$ LHS = $\cos \pi - \sqrt{3} \sin \pi + 1$

$$= -1 - \sqrt{3}(0) + 1$$

$$= 0$$

$$= \text{RHS} \quad \therefore x = \pi \text{ soln}$$

soln: $x = \frac{\pi}{3}, \pi$

1

$$\frac{1-t^2}{1+t^2} - \frac{2\sqrt{3}t}{1+t^2} + 1 = 0$$

applies t-formula

2) $t = \frac{1}{\sqrt{3}}$

$$\tan \frac{x}{2} = \frac{1}{\sqrt{3}}$$

$$x = \frac{\pi}{3}$$

3) includes

$x = \pi$ as

soln.

Question 12

- 2 -

12) b) $I = \int x \sqrt{x-4} dx$

$$I = \int (u+4) \sqrt{u} du$$

$$= \int u^{\frac{1}{2}}(u+4) du$$

$$= \int u^{\frac{3}{2}} + 4u^{\frac{1}{2}} du$$

$$= \frac{2}{5} u^{\frac{5}{2}} + 4 \cdot \frac{2}{3} u^{\frac{3}{2}} + C$$

$$= \frac{2}{5} (x-4)^{\frac{5}{2}} + \frac{8}{3} (x-4)^{\frac{3}{2}} + C$$

$$= \frac{2}{5} \sqrt{(x-4)^5} + \frac{8}{3} \sqrt{(x-4)^3} + C$$

$$\begin{aligned} u &= x-4 \\ \frac{du}{dx} &= 1 \\ du &= dx \\ x &= u+4 \end{aligned}$$

1) $\frac{du}{dx} = 1$

$$x = u+4$$

$$\int u^{\frac{3}{2}} + 4u^{\frac{1}{2}} du$$

or equivalent

2) $\frac{2}{5} u^{\frac{5}{2}} + \frac{8}{3} (x-4)^{\frac{3}{2}} + C$

3) $\frac{2}{5} (x-4)^{\frac{5}{2}} + \frac{8}{3} (x-4)^{\frac{3}{2}} + C$

12) (c) $P(x) = (9x^2-1) \cdot Q(x) + (3x+7)$

$$P(x) = (3x+1)(3x-1) \cdot Q(x) + (3x+7)$$

so $P(-\frac{1}{3}) = 0 + 3(-\frac{1}{3}) + 7$

$$= 6$$

$$\text{remainder} = 6$$

1) correct polynomial

2) uses $P(-\frac{1}{3})$ and correct soln.

Question 12

12 (d) $r(t) = (vt \cos \alpha) \hat{i} + (vt \sin \alpha - \frac{g}{2} t^2) \hat{j}$

i) $v(t) = (v \cos \alpha) \hat{i} + (v \sin \alpha - gt) \hat{j}$

max height at $\dot{y} = 0$

$$v \sin \alpha - gt = 0$$

$$t = \frac{v \sin \alpha}{g}$$

sub $\rightarrow y = v \left(\frac{v \sin \alpha}{g} \right) \sin \alpha - \frac{g}{2} \left(\frac{v \sin \alpha}{g} \right)^2$

$$h = \frac{v^2 \sin^2 \alpha}{g} - \frac{v^2 \sin^2 \alpha}{2g}$$

$$h = \frac{2v^2 \sin^2 \alpha - v^2 \sin^2 \alpha}{2g}$$

$$h = \frac{v^2 \sin^2 \alpha}{2g}$$

II

$$v(t) = v \cos \alpha \hat{i} + (v \sin \alpha - gt) \hat{j}$$

and

$$v \sin \alpha - gt = 0$$

$$t = \frac{v \sin \alpha}{g}$$

2 substitutes t correctly into g and simplifies

12 d) (iii) horizontal distance (x)

at $t = \frac{(2+\sqrt{3})v \sin \alpha}{2g}$

$$d = \frac{v \cos \alpha (2+\sqrt{3})v \sin \alpha}{2g}$$

$$d = \frac{v^2 2 \sin \alpha \cos \alpha (2+\sqrt{3})}{4g}$$

$$d = \frac{v^2 \sin 2\alpha (2+\sqrt{3})}{4g}$$

II

$$\text{dist} = v \cos \alpha \left[\frac{(2+\sqrt{3})v \sin \alpha}{2g} \right]$$

and some correct simplification.

12

If you use this space, clearly indicate which question you are answering.

(d)

(11)

$$\frac{h}{4} = \frac{v^2 \sin^2 \alpha}{8g}$$

$$\text{sub: } \frac{v^2 \sin^2 \alpha}{8g} = vt \sin \alpha - \frac{g}{2} t^2$$

$$\frac{g}{2} t^2 - vt \sin \alpha + \frac{v^2 \sin^2 \alpha}{8g} = 0$$

$$t = \frac{v \sin \alpha \pm \sqrt{v^2 \sin^2 \alpha - 4 \left(\frac{g}{2}\right) \cdot \frac{v^2 \sin^2 \alpha}{8g}}}{\frac{g}{2}}$$

$$t = \frac{v \sin \alpha \pm \sqrt{v^2 \sin^2 \alpha - \frac{v^2 \sin^2 \alpha}{4}}}{\frac{g}{2}}$$

$$t = \frac{v \sin \alpha \pm \sqrt{\frac{3}{4} v^2 \sin^2 \alpha}}{\frac{g}{2}}$$

$$t = \frac{v \sin \alpha \pm \frac{\sqrt{3}}{2} v \sin \alpha}{\frac{g}{2}} \times \frac{2}{2}$$

$$= \frac{2v \sin \alpha + \sqrt{3} v \sin \alpha}{2g}$$

$$t = \frac{(2 + \sqrt{3}) v \sin \alpha}{2g} \quad t > 0$$

need $t > \frac{v \sin \alpha}{g}$, use t -value on way

so $t = \frac{(2 + \sqrt{3}) v \sin \alpha}{2g}$

down
so bigger
value.

1

$$\frac{h}{4} = \frac{1}{4} \left(\frac{v^2 \sin^2 \alpha}{2g} \right)$$

$$4gt^2 - 8gvt \sin \alpha + v^2 \sin^2 \alpha = 0$$

$$gt^2 - 2vt \sin \alpha + \frac{v^2 \sin^2 \alpha}{4g} = 0$$

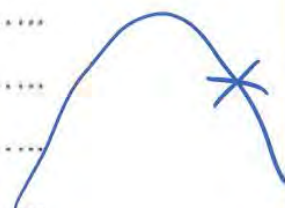
$$\frac{g}{2} t^2 - vt \sin \alpha + \frac{v^2 \sin^2 \alpha}{8g} = 0$$

(or equivalent)

$$vt \sin \alpha - \frac{gt^2}{2} = \frac{h}{4}$$

2. develops quadratic and sub int quad formula

3. selects correct t from quadratic eqn.



Question 13 (15 marks)

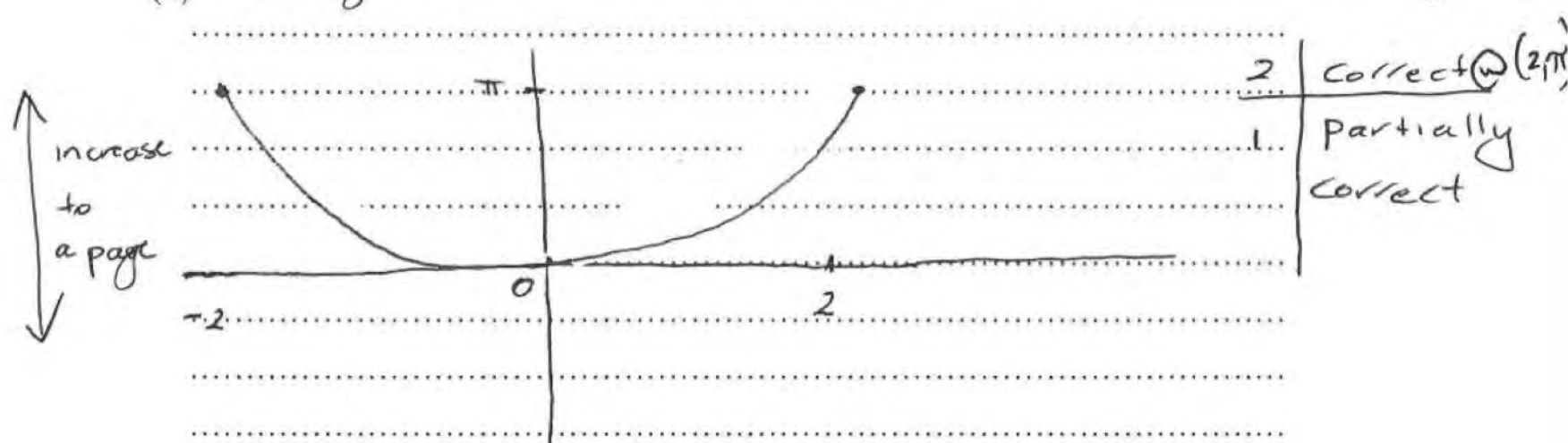
Marks

(a)

(i) $f(-x) = -x \sin^{-1}\left(\frac{-x}{2}\right)$ since $f(x) = f(-x)$
 $= x \sin^{-1}\left(\frac{x}{2}\right)$ the f^n is even
 $f(x) = x \sin^{-1}\left(\frac{x}{2}\right)$

1 r/w

(ii) Change to a blank set of axes with lines 2



$f(2) = 2 \sin^{-1} 1 = \pi$ $f(0) = 0$ $f'(x) = x \times \frac{1/2}{\sqrt{1-(x/2)^2}} + \sin^{-1}\left(\frac{x}{2}\right)$

$= \frac{2x}{2\sqrt{4-x^2}} + \sin^{-1}\left(\frac{x}{2}\right)$
 $= \frac{x}{\sqrt{4-x^2}} + \sin^{-1}\left(\frac{x}{2}\right)$

(b)

When $t = 0$ $T = 20 + 80e^0$
 $= 100^\circ$

$\frac{dT}{dt} = 80 \times (-0.2) e^{-0.2t}$
 $= -0.2 \times 80 e^{-0.2t}$
 $= -0.2(T - 20)$

When $T = 50^\circ$ $\frac{dT}{dt} = -0.2(50^\circ - 20^\circ)$
 $= -6^\circ$

$f'(1) > 0$ $\therefore (0,0)$
 $f'(-1) < 0$ is a min

2	Correct
1	Finding derivative and initial temp

$\therefore$ The water is cooling at 6°C/min

Question 13 continues on page 12

(c)

(i)

$$\cos 3x = 0 \text{ or } \sin 2x = 0$$

$$3x = \frac{\pi}{2}, \frac{3\pi}{2}, \dots \quad 2x = 0, \pi, 2\pi, \dots$$

$$x = \frac{\pi}{6}, \frac{\pi}{2}, \dots \quad x = 0, \frac{\pi}{2}, \pi, \dots$$

$\therefore$ smallest positive solⁿ is $x = \frac{\pi}{6}$

2	multiple solutions for both, showing smallest
1	$\cos 3x = 0$ and $\sin 2x = 0$

(ii)

$$V = \pi \int_0^{\pi/6} \left(\sqrt{\cos(3x) \sin(2x)} \right)^2 dx$$

$$= \pi \int_0^{\pi/6} \cos 3x \sin 2x dx$$

$$= \frac{\pi}{2} \int_0^{\pi/6} (\sin 5x - \sin x) dx \quad *$$

$$= \frac{\pi}{2} \left[-\frac{\cos 5x}{5} + \cos x \right]_0^{\pi/6}$$

$$= \frac{\pi}{2} \left\{ -\frac{1}{5} \cos \frac{5\pi}{6} + \cos \frac{\pi}{6} - \left(-\frac{1}{5} \cos 0 + \cos 0 \right) \right\}$$

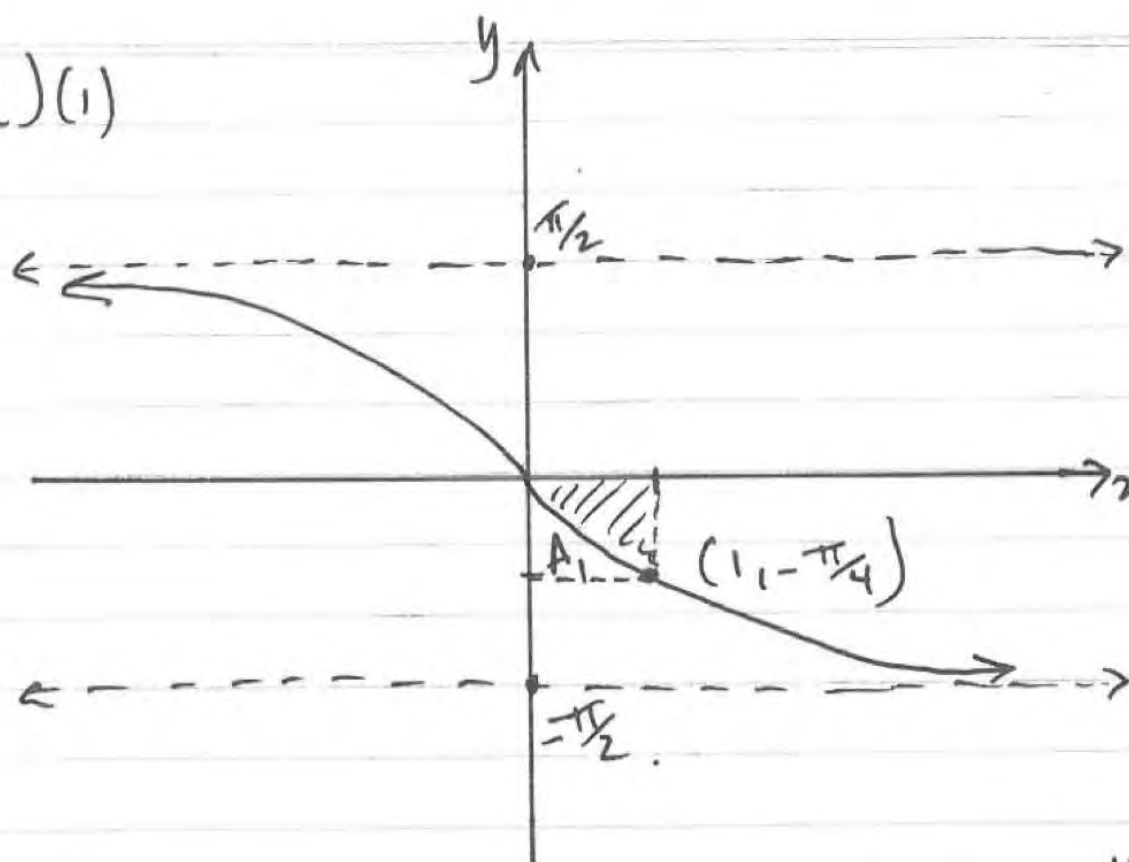
$$= \frac{\pi}{2} \left\{ \frac{\sqrt{3}}{10} + \frac{\sqrt{3}}{2} + \frac{1}{5} - 1 \right\}$$

$$= \frac{\pi}{2} \left\{ \frac{6\sqrt{3}}{10} - \frac{4}{5} \right\}$$

$$= \pi \left\{ \frac{3\sqrt{3}}{10} - \frac{2}{5} \right\} \text{ units}^3$$

3	Collect
2	correct integral expression
1	Getting +0 *

(d)(i)



2	graph with A (1, - $\pi/4$) with scale correct
1	correct shape

$$y = -\tan^{-1}(x)$$

$$x = \tan(-y)$$

$$= -\tan y$$

$$(ii) A = \frac{\pi}{4} - \int_{-\pi/4}^0 -\tan y \, dy$$

$$= \frac{\pi}{4} - \int_{-\pi/4}^0 \frac{-\sin y}{\cos y} dy$$

$$= \frac{\pi}{4} - \left[\ln|\cos y| \right]_{-\pi/4}^0$$

$$= \frac{\pi}{4} - \left[\ln|\cos 0| - \ln|\cos(-\pi/4)| \right]$$

$$= \frac{\pi}{4} - \left[\ln 1 - \ln \frac{1}{\sqrt{2}} \right]$$

$$= \frac{\pi}{4} - \left[0 + \ln \sqrt{2} \right]$$

$$= \left(\frac{\pi}{4} - \frac{1}{2} \ln 2 \right) \text{ units}^2$$

3	correct evaluation
---	-----------------------

2	progress to to to calculate A
---	--

1	$\frac{\pi}{4} - A_1$
---	-----------------------

AND FYI $\ln \frac{1}{\sqrt{2}}$ is a negative number!

NOTE: Common error were incorrectly using absolute value brackets for A_1 , and consequently having incorrect sign.

Question 14 (16 marks)

Marks

(a)

3

When $t=0$ $v=0$

$$\frac{dv}{dt} = (v+1)^{1/2}$$

$$\frac{dv}{(v+1)^{1/2}} = dt$$

$$\int \frac{1}{\sqrt{v+1}} dv = t + C$$

$$2(v+1)^{1/2} = t + C \quad *$$

When $t=0$ $v=0 \therefore C=2$

$$\therefore 2\sqrt{v+1} = t + 2$$

When $t=1$ $\sqrt{v+1} = \frac{3}{2}$

$$v+1 = \frac{9}{4}$$

$$v = \frac{5}{4} \text{ m/s}$$

3	Correct
2	Working in progress
1	Getting to *

(b)

1 r/w

(i) $P = 400$ penguins

Question 14 continues on page 18

Question 14
Part (b) continued

Marks

3

$$(ii) \frac{dP}{dt} = 0.001P(400-P)$$

$$\int \frac{dP}{P(400-P)} = \int 0.001 dt$$

$$\frac{1}{400} \int \left(\frac{1}{P} + \frac{1}{400-P} \right) dP = 0.001t + C$$

$$\frac{1}{400} [\ln|P| - \ln|400-P|] = 0.001t + C \quad **$$

$$\text{When } t=0 \quad P=50$$

$$\frac{1}{400} \left[\ln \frac{50}{350} \right] = C$$

$$C = \frac{1}{400} \times -\ln 7$$

$$= \frac{-\ln 7}{400}$$

$$\frac{1}{400} \ln \left| \frac{P}{400-P} \right| = 0.001t - \frac{\ln 7}{400}$$

$$\ln \left| \frac{P}{400-P} \right| = 0.4t - \ln 7 \quad *$$

$$\text{When } P=200$$

$$0.4t - \ln 7 = \ln \left| \frac{200}{200} \right|$$

$$0.4t = \ln 7$$

$$t = 4.864775373 \text{ years}$$

$$\approx 4 \text{ years } 10 \text{ months}$$

3	correct
2	Gets to * or equivalent.
1	Gets to ** or equivalent

Question 14 continues on page 19

(c)

(i)

2

Side length of the square is $2r$

$$\vec{AP} = \vec{AO} + \vec{OP}$$

$$\vec{AP} \cdot \vec{AP} = (\vec{AO} + \vec{OP}) \cdot (\vec{AO} + \vec{OP})$$

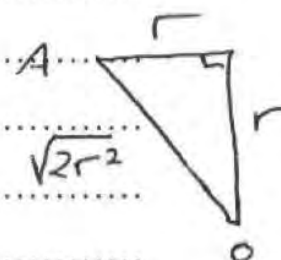
$$= \vec{AO} \cdot \vec{AO} + 2\vec{AO} \cdot \vec{OP} + \vec{OP} \cdot \vec{OP}$$

$$* = |\vec{AO}|^2 + |\vec{OP}|^2 - 2\vec{OA} \cdot \vec{OP}$$

$$= (\sqrt{2}r)^2 + r^2 - 2\vec{OA} \cdot \vec{OP}$$

$$= 2r^2 + r^2 - 2\vec{OA} \cdot \vec{OP}$$

$$= 3r^2 - 2\vec{OP} \cdot \vec{OA} \quad \text{As req'd}$$



2	Correct
1	Arrives at
	*

Question 14
Part (c) continued

Marks

2

(ii)

$$AP^2 = \vec{AP} \cdot \vec{AP} = 3r^2 - 2\vec{OP} \cdot \vec{OA}$$

Similarly

$$BP^2 = 3r^2 - 2\vec{OP} \cdot \vec{OB}$$

$$CP^2 = 3r^2 - 2\vec{OP} \cdot \vec{OC}$$

$$DP^2 = 3r^2 - 2\vec{OP} \cdot \vec{OD}$$

Adding

$$\begin{aligned} AP^2 + BP^2 + CP^2 + DP^2 &= 4 \times 3r^2 - 2\vec{OP} \cdot (\vec{OA} + \vec{OB} + \vec{OC} + \vec{OD}) \\ &= 12r^2 - 2\vec{OP} \cdot \vec{0} \\ &= 12r^2 \end{aligned}$$

2	Correct
1	Set up and link to (i)

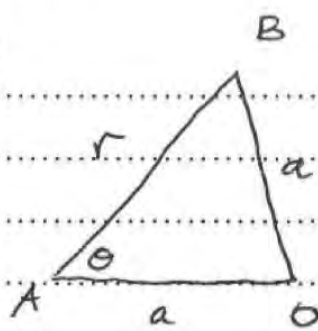
Question 14 continued

Marks

(d)

(i)

2



$$AB = r$$

$$AO = a$$

$$OB = a$$

$$\cos \theta = \frac{a^2 + r^2 - a^2}{2ar}$$

$$= \frac{r^2}{2ar}$$

$$\cos \theta = \frac{r}{2a}$$

$$\therefore r = 2a \cos \theta$$

2	Both correct
1	Correct

$$L = r \times 2\theta$$

$$= 2a \cos \theta \times 2\theta$$

$$L = 4a\theta \cos \theta$$

Question 14 continued

Marks

Part (d) continued.

(ii)

3

$$\frac{dL}{d\theta} = 4a\theta \times -\sin\theta + \cos\theta \times 4a$$

$$= -4a\theta \sin\theta + 4a \cos\theta$$

$$\frac{d^2L}{d\theta^2} = -4a\theta \cos\theta + \sin\theta \times -4a - 4a \sin\theta$$

$$= -4a\theta \cos\theta - 8a \sin\theta$$

When $\frac{dL}{d\theta} = 0$ $4a\theta \sin\theta = 4a \cos\theta$

$$\theta \tan\theta = 1$$

$$\theta = \frac{1}{\tan\theta}$$

$$= \cot\theta$$

When $\theta = \cot\theta$ $\frac{d^2L}{d\theta^2} = -4a \cot\theta \cos\theta - 8a \sin\theta$

Since $0 < \theta < \frac{\pi}{2}$ $\cot\theta, \sin\theta \cos\theta > 0$

$$\therefore \frac{d^2L}{d\theta^2} = -4a(\cot\theta \cos\theta + 2\sin\theta) < 0$$

$\therefore \theta = \cot\theta$ will produce a maximum L

3	Correct
2	Finds derivatives shows $\theta = \cot\theta$ is a stationary pt
1	Finds derivatives

END OF PAPER